

Material-Systems Intelligence via Data, Science, and Data Science

Christopher Rudolf^{1*}, James Amarel⁴, James Thomas¹, Athanasios Iliopoulos², John Michopoulos², Leslie Smith³

US Naval Research Laboratory

¹Multifunctional Materials Branch

²Center for Materials Physics and Technology

³Navy Center for Applied Research in Artificial Intelligence

⁴NRC Postdoctoral Fellow

4555 Overlook Ave. SW, Washington, DC 20375

Abstract

Structural health monitoring (SHM) increasingly leverages deep learning to model nonlinear relationships between sensor signals and structural condition, but training such models typically requires large, well-labeled datasets. This paper centers on an integrated software–hardware system designed to generate high-quality training data for machine learning–based assessment of material damage in thin metal plates using active sensing with Lamb waves. The system comprises two coordinated data-generation pathways. First, an automated system with robotic control induces and parameterizes damage via controlled contact loads, while a network of piezoelectric wafer active sensors (PWAS) actuates and records Lamb-wave responses. Purpose-built control and acquisition software synchronizes actuation, sensing, and robotic motion; captures raw waveforms and environmental metadata; and assigns structured labels to support supervised learning. Second, a computational modeling component synthesizes signals under corresponding damage scenarios to augment and diversify the training set, aligning simulation parameters with experimental conditions to improve coverage across wave modes and boundary interactions. Because Lamb waves propagate over long distances yet are dispersive and multimodal (symmetric and antisymmetric), their interactions with damage sites, boundaries, and surface attachments produce complex, noisy signals. The system explicitly exploits these characteristics by varying excitation patterns and damage configurations, yielding material-response-aware datasets suitable for training artificial neural networks (ANNs) for detection, localization, and characterization. The paper details the system’s architecture, control software, calibration and labeling procedures, and data schema, and uses the resulting datasets to demonstrate accurate and reliable quantification of material condition within this SHM context.

1. Introduction

Structural-Health Monitoring (SHM) is a damage identification and quantification discipline that enables observations over time to recognize and measure damage-sensitive features and determine the current state of health. This information can be used to estimate performance margin and system integrity in the presence of damage and during extreme events. SHM methods are often classified into two groups: model-based approaches, which rely on mathematical or physical models to predict behavior, and data-driven approaches, which use measurements acquired directly from the material or structure [1]. The goal of this research is to pursue a hybrid strategy in which experimentally generated data provides the majority of training samples, supplemented with computationally modeled data to account for previously unlearned

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- 1) Experimental Data Generation for Guided-Wave Damage Presence Detection: A Scalable SHM Approach
- 2) A Methods Framework for Building Robust Machine Learning Training Data for Guided-Wave Damage Presence Detection
- 3) Autonomous Experimental Platform for High-Throughput Lamb-Wave Data Generation to Train Damage-Detection Models
- 4) The Sound of Damage: Hardware-Software Synergy for Data-Driven Lamb-Wave Diagnostics

environmental, operational factors, and damage conditions. This approach enables learning material-specific complexities while retaining the generalization necessary for broader use. To address the practical need for large, well-labeled datasets, we develop an integrated software–hardware system that couples active Lamb-wave sensing with automated experimental data generation and coordinated computational synthesis, producing training data tailored to machine learning (ML) models for damage assessment.

To gather the necessary datasets, sensing must provide sufficient information to infer the structural state. Sensing systems can be active, passive, or a combination thereof. This research focuses on active sensing in which a prescribed excitation signal interacts with the structure and is altered when damage or a defect is present. Elastic Lamb (plate) waves are employed to sense material damage in the form of voids, cracks, and corrosion. Lamb waves can be generated using commercial piezoelectric wafer active sensors (PWAS). PWAS sensors (often referred to as PZTs) leverage the piezoelectric effect, which couples mechanical stress and electrical charge, allowing them to function as both actuators and sensors [2]. Lamb waves can travel large distances (several meters) with low attenuation in metal and composite structures. Multiple Lamb-wave modes are possible, including symmetric (S) modes in which particle motion is primarily in-plane and parallel to the propagation direction, and antisymmetric (A) modes in which particle motion is primarily out-of-plane and perpendicular to the propagation direction. Both symmetric and antisymmetric modes propagate through the full plate thickness, and the excitation frequency should be tailored to achieve the desired mode(s). The presence of multiple modes (e.g., S₀, A₀, S₁, A₁, ...) significantly complicates reflections and scattering from damage sites and boundaries in the material [3, 4]. The typically intractable waveform-structure interactions, combined with mode conversions and commonly applied signal transformations, yield complex, noisy data that is well-suited for developing material response-aware ML algorithms.

A range of classical ML methods has been studied for SHM using guided-wave signals, including decision trees, linear regression, support vector machines, and k-nearest neighbors [5, 6]. While these methods can perform damage/no-damage classification, Artificial Neural Networks (ANNs) offer greater flexibility for higher-order tasks. Damage state identification is typically framed hierarchically: 1) detection, 2) localization, 3) classification of damage severity, and 4) prognosis [7]. With ML, each level is attainable in principle, but only with sufficient training data; moreover, data requirements typically grow substantially with each successive level. Because it is rarely cost-effective to obtain experimental specimens that represent every possible damage scenario, computational modeling is frequently used to augment training sets.

Prior work adapting ANNs to materials problems often repurposes networks optimized for vision and speech recognition. This adaptation typically requires preprocessing and feature extraction of raw 1D time-domain signals to produce inputs amenable to off-the-shelf ANNs, many of which expect 2D inputs [5, 6, 8]. Another common approach employs 1D convolutional neural networks with sensor fusion across multiple channels [9, 10]. Similarly, we design ANNs to accept direct inputs of Lamb-wave signals; in contrast to much previous work, we explicitly ensure that the architecture respects time-translation symmetry, which is inherent to the physics of wave propagation.

Theoretical analyses of wave propagation in complex elastic structures are mathematically challenging even for simple geometries (e.g., the semi-infinite half-plane). Finite element analyses of structural elastodynamics are feasible, but accurate predictions generally require

element sizes $o(\lambda/10)$ and time steps $o(T/10)$ to resolve transient deformations associated with kilohertz-to-megahertz excitation and phase velocities exceeding $>10^3$ m/s in metals and composites. Complex geometries, interacting boundaries, dispersion, interference, attenuation, and the effectively limitless variability of damage characteristics render exhaustive calculations and labeling for all scenarios computationally prohibitive. These factors limit current SHM capabilities for reliably locating and characterizing damage across diverse materials and structural components. The framework proposed here is a first step toward overcoming these limitations by determining how to implement neural networks optimized for material–structure applications and by establishing a reproducible pathway for dataset generation. Because it is impractical to deterministically model all deleterious factors affecting structural response, we envision a framework that learns features relevant to damage attributes while de-emphasizing confounding influences (e.g., joint dissimilarities, interfaces, internal sub-component boundaries). This paper addresses two major gaps in ML for SHM: 1) over-reliance on computationally generated data, which can be too pristine and lead to inaccurate testing on real systems, and 2) use of ML/ANN algorithms developed for image analysis applied without adaptation to time-series sensor data from physics-based systems. Our integrated software–hardware data-generation system directly targets gap (1), while our ANN design addresses gap (2), together supporting accurate, scalable damage quantification in plate-like structures.

2. Methodologies

In this work, a 1.2 mm thick plate of Al6061-T6 with dimensions 610 mm x 610 mm was utilized as the test bed. Based on the calculated group and phase velocity dispersion curves, a 300 kHz, 5 volt peak-to-peak, 5 cycle, Hanning windowed sine wave with a 100 ms tone burst period was chosen. Using a Hanning window on the signal narrows the signal bandwidth and concentrates the input energy at a central frequency, which helps to alleviate some of the dispersion that may occur. Additional methodology details are provided in the following subsections.

2.1 Structural Health Monitoring System – Lamb Wave Sensing with piezoelectric wafer active sensors (PWAS)

Lamb waves are an elastic form of ultrasonic guided waves that propagate parallel to the surface of a thin plate. They are termed guided waves because of the way they remain confined to the boundaries of the plate via the free parallel upper and lower surfaces of the plate [11]. Lamb waves are dispersive, meaning their phase velocity, c_p , depends on frequency. Multiple simultaneous wave modes, both symmetric and antisymmetric, are generated depending on the frequency, host material properties, and thickness. These are determined by the Rayleigh-Lamb equations in an isotropic plate, given by Equation 1 for symmetric modes and Equation 2 for antisymmetric modes:

$$\frac{\tan(qh)}{\tan(ph)} = -\frac{4k^2 qp}{(k^2 - q^2)^2} \quad \text{Equation 1}$$

$$\frac{\tan(qh)}{\tan(ph)} = -\frac{(k^2 - q^2)^2}{4k^2 qp} \quad \text{Equation 2}$$

where h is the plate thickness, k is the wave number. The terms p , q , and k are defined as:

$$p^2 = \frac{\omega^2}{c_L^2} - k^2$$

$$q^2 = \frac{\omega^2}{c_S^2} - k^2$$

$$k = \frac{\omega}{c_p}$$

where c_L is the longitudinal wave velocity, c_S is the shear (transverse) wave velocity, ω is the wave angular frequency, and c_p is the phase velocity. The longitudinal and shear wave velocities are determined by:

$$c_L = \sqrt{\frac{E(1-\nu)}{\rho(1+\nu)(1-2\nu)}}$$

$$c_S = \sqrt{\frac{E}{2\rho(1+\nu)}}$$

where E is the Young's modulus, ν is Poisson's ratio, and ρ is the density. Numerically solving Equations 1 and 2 leads to the phase and group velocities that can be plotted against the frequency-thickness product, resulting in the dispersion curves. In this research, the dispersion curves were solved for and plotted using Matlab and Dispersion Calculator 3.1 code [12]. The results are shown in [Figure 1](#) and [Figure 2](#). The dispersion curves show that at the frequency thickness product (fd) of 0.36 MHz*mm utilized in this work (frequency = 300 kHz, plate thickness = 1.2mm), two fundamental modes will propagate (i.e., S0 and A0).

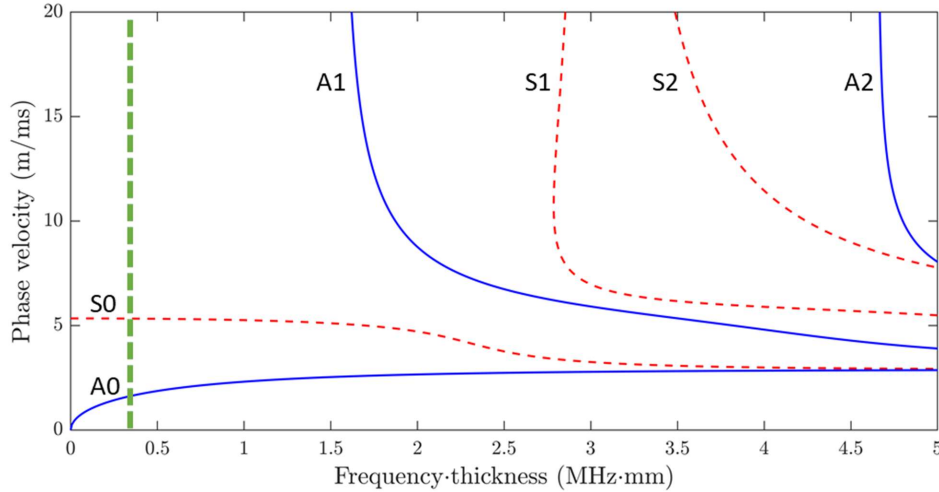


Figure 1: Phase velocity dispersion curve for an Aluminum 6061T6 plate. Dotted vertical line indicates $fd = 0.36 \text{ MHz*mm}$.

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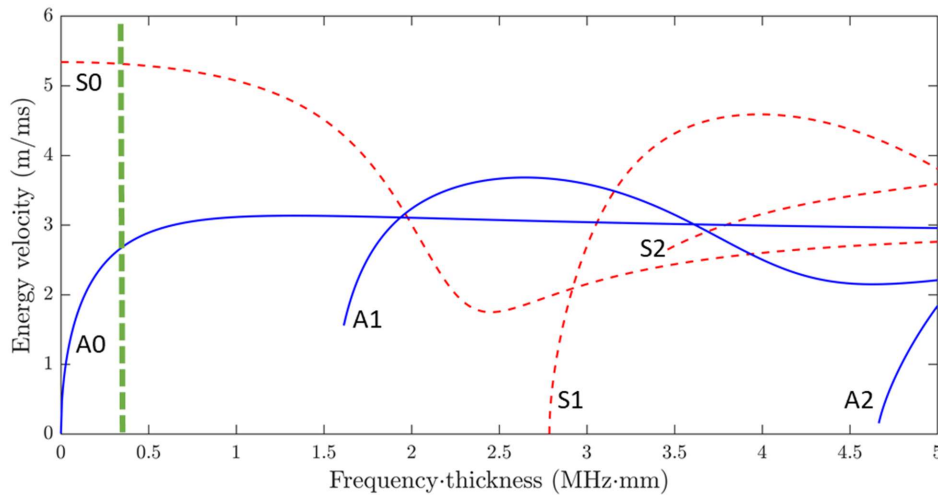


Figure 2: Group (energy) velocity dispersion curve for an Aluminum 6061T6 plate. Dotted vertical line indicates $fd = 0.36 \text{ MHz} \cdot \text{mm}$.

Piezoelectric wafer active sensors (PWAS) are a common choice for generating Lamb waves. They operate on the piezoelectric principle and achieve direct transduction of electric energy into elastic energy and vice versa (i.e., an alternating voltage applied to the PWAS terminals produces a radial oscillatory expansion and contraction of the PWAS material) which makes them excellent excitation sources as well as sensors [13]. In Lamb wave applications, PWAS excite and sense the Lamb waves in the structure directly through in-plane strain coupling [2]. PWAS with a diameter of 9.5 mm and thickness of 1 mm (Model 63, APC International) are utilized by epoxy bonding to the metal plates according to the schematic shown in [Figure 3](#). The epoxy used for bonding was Loctite E-60HP structural adhesive mixed with 0.05 mm diameter spheres of yttria stabilized zirconia (YSZ) at a 1.5% by volume sphere/epoxy ratio. The use of the spheres ensured a consistent, repeatable bonding thickness between all PWAS sensors. After bonding and soldering the wire leads, a layer of protectant is applied (Micromeritics M-Coat A polyurethane) to the PWAS for environmental protection. Four PWAS are symmetrically attached to the plate with vertical and horizontal separations of 350 mm (x,y locations of (130,130), (130,480), (480,130), and (480,480)). For each damage scenario, each of the four sensors is utilized to send the generated Lamb waves while the other three act as receivers.

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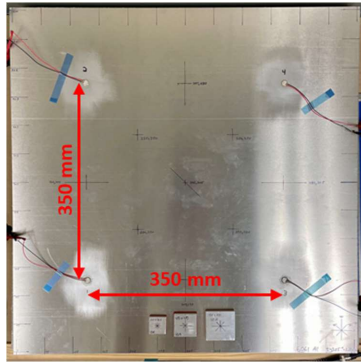


Figure 3: Figure of test plate with PWAS bonded.

2.2 Simulated Damage using Contact Loads

It is not feasible to have thousands of metal plates with various cracks/damage. Therefore, a method of using “simulated” damage via contact loads was devised in order to experimentally generate training data for the ML algorithms. Previous work has shown methods of simulating damage to plates using contact loads including bonded additional masses that simulate damage by changing the local stiffness and scattering the wave similar to what would be seen when delamination occurs [8, 14]. Other methods include using metal blocks coupled with ultrasonic gel that attenuated the wave signals [15], applying absorbing mud/putty that causes attenuation and phase shifting of the waves [10, 16], using a C-clamp to change the stiffness and inertia of the structure in a localized area causing the incident Lamb waves to be scattered from it similarly to a crack [17], and aluminum tape adhered to a surface which causes an amplitude change of the received signal [18, 19]. While these previous studies showed the potential of using a contact load for simulating damage, the contact loads utilized were permanently or semi-permanently bonded which limits the ability to automate the generation of damage location data over the test plate. In this research, we aimed to autonomously move the contact load which would require a temporary coupling method that would not leave any residue when moved. A 40 mm x 40 mm square of aluminum was utilized as the contact load and a post was added to it to enable pick and place using a two-finger gripper. Couplants were investigated including ultrasonic gel, Eco-Flex 00-20 platinum cure silicone, and silicone sheets (1/8-inch-thick) of durometers 10A, 20A, 30A, and 40A. The silicone sheets were coupled to the aluminum squares via double sided tape.

2.3 Automated Experimental Platform

A platform for autonomously generating labelled location data, shown in [Figure 4](#), was assembled and included a robotic gantry, a signal generator, an oscilloscope, and a computer running a custom Matlab program for both controlling the gantry movements and signaling and acquiring the data from the oscilloscope. An XYZ gantry (Igus inc.) was designed and assembled for the purpose of moving the contact load around the plate to simulate damage locations. The gantry has a working area of 650 mm x 650 mm x 150 mm in the x, y, and z axis, respectively. Attached to the z-axis is a strain wave gear and a Schunk 2-finger gripper (Model EGP64) that can pick the contact load up, rotate it to a desired orientation, and place it back on the plate.

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The Lamb wave is generated using an Agilent 33500B waveform generator and is sent via one of the PWAS. The signal is received at the remaining three PWAS via a Tektronix MSO4104B mixed signal oscilloscope. The oscilloscope acquired a 400 μ s signal length sampled at 25 MS/s across four channels (one send that triggered the acquisition and three receive signals). Each set of experiments included 3 baseline signals taken at the start of the testing and 3 at the end. The Matlab program synchronized the automation to trigger when and where to move the contact load, when to acquire the signals via the oscilloscope, and signal when acquisition was completed to move to the next location. On average, each contact load position took 12 seconds including movement and acquisition time which enabled 300 locations to be acquired per hour. To put this in perspective, manually performing these experiments resulted in about 30 locations being acquired in an hour.

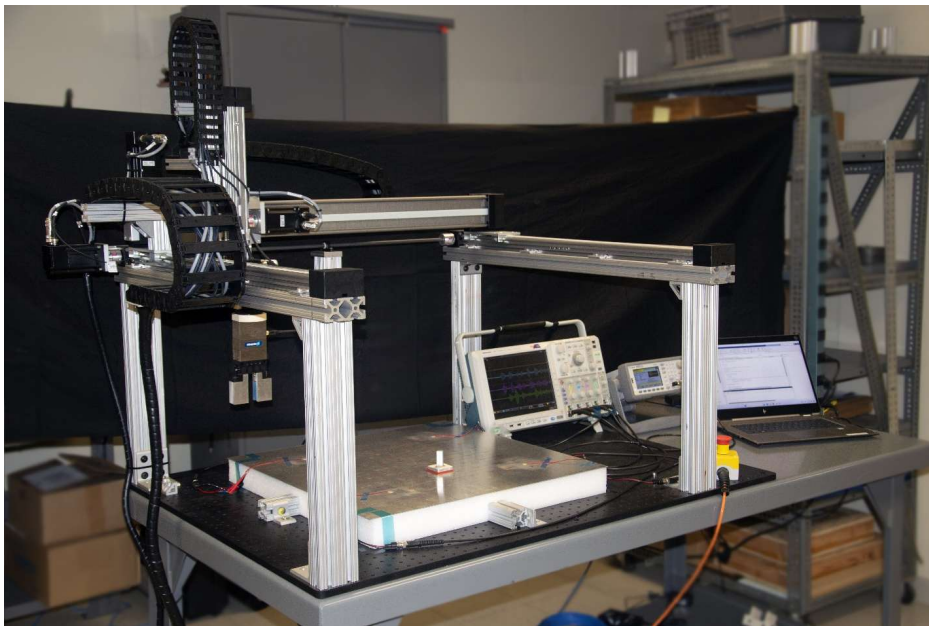


Figure 4: Automated experimental platform.

Initial, manually acquired datasets contained 290 data points per set, taking approximately 10 hours to acquire. Autonomously acquired datasets contained up to 10,428 data points, taking approximately 35 hours to acquire. Importantly, the automated acquisition was set-and-forget and only took about ten minutes of personnel involvement for setup. The size of the datasets that were experimentally acquired rivals computationally simulated dataset sizes but with the benefit of real complexities that are not realized in simulations. For example, the rolling direction on the aluminum plate caused a slight time variation in the vertical vs horizontal sensor pathways even though they are both separated by the same 350 mm distance. The experimentally acquired sensor data is also ideal for validating/calibrating computational models or digital twins. For

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example, the acquired baseline signals can be used to accurately model the bonding conditions between the PWAS and the plate.

2.4 Computational Data Generation

To obtain an even richer set of data we developed a computational framework for performing modeling and simulation on arbitrarily sized square plates with a crack and four piezoelectric transducers. The framework was incorporated into (a) a Graphical User Interface (GUI) application, for exploring different parametric combinations, and also (b) into a combinatorial parametric sweep simulation model appropriate for generating resulting waveforms in a batch. For both of these cases we used the COMSOL Multiphysics FEA software.

The core of the computational framework operates within the principles of solid mechanics and implements the usual equation of motion together with the material constitutive law:

$$\rho \frac{\partial^2 \mathbf{u}}{\partial t^2} = \nabla \cdot \boldsymbol{\sigma},$$

where $\mathbf{u}$ is the displacement vector, $\boldsymbol{\sigma}$ is the stress tensor, and ρ is the density. In the current implementation we assume linear elastic material with a constitutive law $\boldsymbol{\sigma} = \mathbf{C} : \boldsymbol{\varepsilon}$, with $\mathbf{C}$ the stiffness tensor and $\boldsymbol{\varepsilon}$ the strain tensor. We have applied the small strain approximation and therefore strain relates to displacement through:

$$\boldsymbol{\varepsilon} = \frac{1}{2} [(\nabla \mathbf{u})^T + \nabla \mathbf{u}]$$

Because we are using the equation of motion, all the relevant wave mechanics are present, and this results in the natural manifestation of Lamb waves, without the need for any further assumptions.

To achieve data generation for training, the framework is parametrized on several characteristic quantities. These quantities include the plate size, its thickness, the crack length or contact load size and location, as well as its orientation. The dynamic simulation includes the same excitation wave parameters used in the experimentation for generating the Lamb waves. The readout transducers were modeled to match the PWAS used in experimentation (i.e., size and with piezoelectric structural material properties given by the manufacturer) and the readout implemented as a trace of the strain tensor. The trace of the strain sensor maps one-to-one with the piezoelectric voltage change when stressed. The simulation was run with a recording length of 400 μ s and a sampling rate of 25 MHz to match the experimental acquisition. The simulation capability has been parameterized to generate data for training machine-learning models. The generation involves either the time-series of the readout sensors, or sequences of floating-point accuracy images that represent the full wave dynamics on the top surface of the plate.

The GUI application (named LaWaS) was developed using the COMSOL compiler. The benefit of using this approach is that the application is encapsulated in a single executable that links to the COMSOL runtime which includes the software logic for pre-processing, finite element analysis and post-processing steps. The software can be operated by persons that are not familiar with FEA. A screenshot of the application, together with the description of its basic functions are

shown in [Figure 5](#)Figure 5. An example solution showing the displacement vector and wave propagation at four different snapshots is shown in [Figure 6](#)Figure 6.

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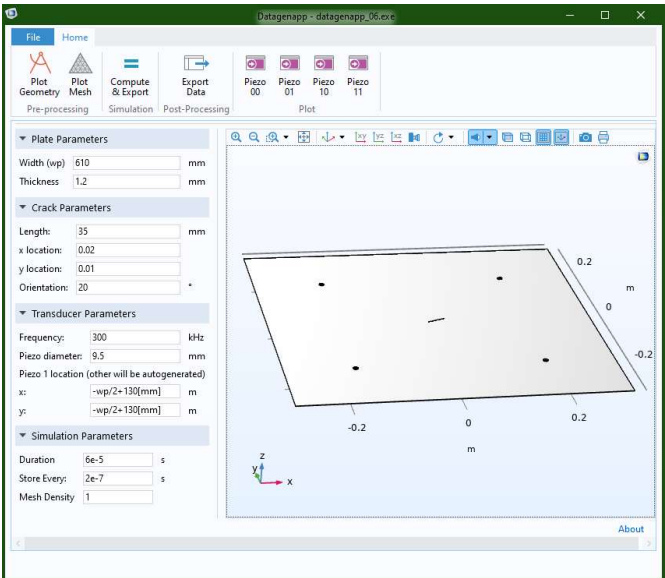


Figure 5: The developed software application (LaWaS) interface for computing the time histories of four piezoelectric transducer output. The application includes functionality for plotting various aspects of the model, executing the analysis and exporting the waveform data represented by the buttons (these functions are located in the ribbon). The user can modify the self-explanatory parameters in the pane on the left, while the geometry, mesh and results can be displayed in the graphical pane on the right.



Figure 6: Example results showing the displacement magnitude for four timepoints

Although the GUI application can be used to generate data for training, it requires human attendance that does not allow for scaling to more than a few hundred simulations, without significant manual labor. To enable massive generation of datasets we repurposed the framework in a manner that the analysis can be performed combinatorically for a selected set of discrete values of the crack location, the crack length, and the crack orientation. The framework was implemented as a parametric batch sweep in COMSOL allowing for solving multiple simulations in parallel. The framework can be deployed in clusters and allow for the generation

of massive data sets in a short amount of time. For the purposes of the current project we deployed it on a 64 core AMD system with 2Tb of memory, running 32 simulations at the same time. The generation of 1,100 datasets for the three receiving transducers completed in about three weeks.

2.5 Machine Learning Approach

Summary of the ANN algorithm and approaches used for this paper, Possibly a comparison with other ML methods?

The artificial neural network (ANN) developed for this work is a one-dimensional temporal convolutional model designed to infer either the presence or spatial location of damage in a thin aluminum plate using Lamb-wave responses. The experimental setup employed four PWAS mounted in a square pattern on the plate, each operating sequentially as an actuator while the remaining three served as receivers. This procedure produced, for each damage state, a full 4×4 matrix of sender-receiver measurements, forming a multi-channel input tensor that captured all propagation paths simultaneously. Prior to training, the raw signals were band-pass filtered, downsampled, and time-trimmed. The resulting processed traces were concatenated across the 16 propagation paths so that the network could jointly analyze all four excitation events in a causally consistent manner. The ANN comprised a series of temporal convolutional layers with nonlinear activations and normalization, enabling it to capture both early-arrival and reflected wave components that encode spatial information about damage.

A summary of the machine learning architecture is provided here. For in-depth details of the approach and results of damage detection and localization compared between three different architectures, we refer you to our prior publication [20]. The architecture employed was a convolutional neural network (CNN) with six blocks. The first five blocks contained convolutional layers with kernel sizes equal to the input sequence length, symmetrically padded to maintain sequence integrity. The receiver and sender indices were treated as channels, and nonlinearities were omitted in the first block. Internal layers included skip connections, LayerNorm [21], and the swish [22] activation function. The final block utilized a dense layer to process boundary effects and output the damage detection results. To enhance the model's generalization and robustness, symmetry constraints were incorporated into the architecture. Specifically, the best performing network was designed to be approximately equivariant, allowing it to capture weak symmetry-breaking effects caused by material anisotropy, boundary irregularities, and non-ideal aspects of Lamb wave generation. This approach improved the model's ability to generalize across different damage states and reduced the generalization gap compared to ordinary CNNs. The training process involved the Adam optimizer [23] with a OneCycle [24] learning schedule. The dataset was split into training and testing sets with an 80%/20% ratio. This architecture successfully learned to map complex, multimodal Lamb-wave time-series data directly to spatial damage coordinates without relying on manual feature extraction or explicit physical modeling, providing a robust and data-driven framework for real-time structural health monitoring.

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3. Experiments

Table of the experimental datasets used for training (add computational if any used)

To mitigate noise in both measurement and baseline acquisition, each damaged signal was paired with six baseline-subtracted variants, expanding the training dataset accordingly. During inference, predictions from these augmented input-target pairs were ensemble-averaged to improve robustness.

This section will need further writeup once we determine if we are using all 4 datasets in our results/discussions.

Table 1: Experimental dataset attributes.

Dataset ID	CL used	Location Spacing	# Locations	# Baselines per send PWAS	Total signal acquisitions
AT1 (C5)	40 mm	25 mm	121	0	484
AT2 (C6)	40 mm	25 mm	121	6	508
AT3 (C7)	40 mm	5 mm	2601	6	10,428
AT4 (C8)	40 mm	5 mm	2601	6	10,428

All experiments were performed on the same 610 mm x 610 mm x 1.2 mm Al6061 plate. The four PWAS were symmetrically bonded to the plate at the following locations (in mm) with the lower-left corner of the plate being 0,0: PWAS 1 (130,130); PWAS 2 (130,480); PWAS 3 (480,130); PWAS 4 (480,480). All location points were within the bounds of the PWAS and begin at location (180,180) and end at (430,430).

4. Results and Discussion

4.1 Determination of Couplant for Contact Loads

The parametric study on contact load characteristic effects on time-amplitude signal response showed that using experimentally “simulated” damage results in attenuation of the magnitude of voltage response, as shown in [Figure 7](#). It was immediately evident that a couplant between the contact load and metal plate is required to cause an attenuation of the signal. A list of the couplants examined is provided in [Table 2](#). Initial testing has not shown a reflection of the signals from the contact load-Lamb wave interactions as would be observed from an actual crack. The attenuation observed more so mimics what would be seen if corrosion were present on a surface. The wave mode most attenuated in our signal was the A_0 mode, as shown in [Figure 7](#). This is because the A_0 mode has a shorter wavelength and has previously been shown to have greater sensitivity to small-size damage as well as surface damage [25, 26]. The S_0 mode was also attenuated but at a much lower level. The faster propagation speed and less dispersion of the S_0 mode make it better at detecting damage far away from the excitation signal as well as internal defects or deep cracks. The multimode nature of Lamb waves therefore provides for excellent detection of damage both on the surface as well as internal to a system of interest.

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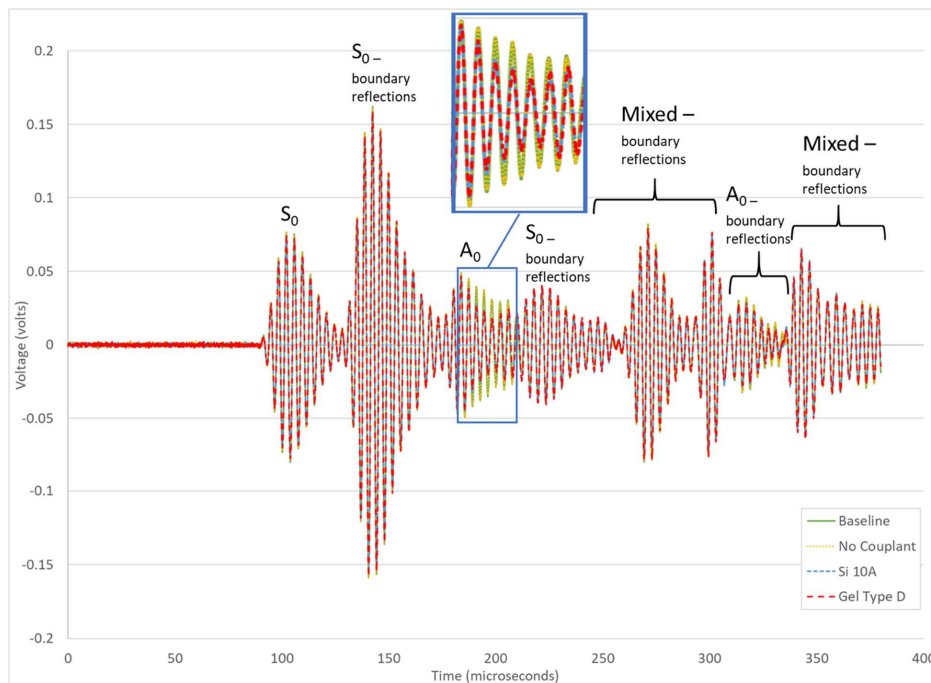


Figure 7: Experimentally acquired PWAS amplitude-time signals comparing the attenuation resulting from different couplants used between the contact load and the metal plate. The different wave modes are labelled as follows: Symmetric mode 0 (S_0), Antisymmetric mode 0 (A_0), and boundary reflections. Figure inset is an expanded view of the incident A_0 signal.

Table 2 Comparison of percent attenuation for different couplants tested (calculated from the incident A_0 signal – inset of [Figure 7](#)).

Couplant Used	Percent Attenuation
No couplant	3.44 %
Silicone sheet – 40A durometer	4.88 %
Silicone sheet – 30 A durometer	7.26 %
EcoFlex 00-20	26.12 %
Silicone sheet – 20 A durometer	29.89 %
Silicone sheet – 10 A durometer	32.81 %
Ultrasonic Couplant Gel Type D	35.08 %

Of the couplants used, ultrasonic gel type-D resulted in the greatest attenuation but required cleaning of the contact load and plate between each location. Cleaning of the plate between locations would have defeated the purpose of automating the experimental data collection.

Silicone sheets, 1/8-inch-thick, with stiffness durometers ranging from 40A (stiffest) to 10A (softest), all attenuated the signal, but the 10A durometer attenuated the A_0 signal most. Therefore, the contact loads used for the autonomous generation of datasets utilized a sheet of 10A durometer silicone as the couplant adhered to the aluminum contact load.

4.2 Experimental Data Size

If Leslie and/or James can run a comparison on the results from AT1+2 vs AT3+4, we could include a discussion on the impact of the size of the datasets for ML accuracy. I found that AT1/AT2 were sufficient for detection but could not support localization with typical errors smaller than the load width. Using both AT3 and AT4 to train a localizer gave marginally better results than using just AT3; to best make use of both AT3 and AT4, one needs a data fusion strategy that overcomes baseline drift.

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4.3 Machine Learning Data Fusion

This section can be renamed depending on what is included (maybe "Data Preparation for Machine Learning" or "Data Fusion"). Because this is a paper focused on the method, perhaps the focus can be on what was tried and what worked/didn't work in terms of data preparation and how that had an effect on the ML Results (compute time, accuracy, etc.). If you have it, maybe including something on data fusion with the computational simulations data as well?

Effective data fusion requires overcoming the sim-to-real gap. One must render commensurate the two signal types, which are not only sampled at different frequencies, but also contain different physics. Ultimately, our previously published model was found to capitalize on real world information sources not captured by the simulation, including effects due to boundaries and wave-mixing. Additionally, the simulation data did not reflect variability in actuator bonding.

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4.4 Damage Detection and Localization

Recent machine learning results demonstrated highly accurate damage detection and precise localization [20]. All three network variants—ordinary, exactly equivariant, and approximately equivariant—achieved detection accuracies exceeding 99%, with the symmetry-aware models converging more rapidly during training. For localization, the approximately equivariant network achieved the best performance, with a mean distance error of 2.58 ± 0.12 mm, compared to 2.93 ± 0.11 mm and 2.98 ± 0.14 mm for the exactly equivariant and ordinary models, respectively. Nearly all test predictions (99.9%) were within 13.4 mm of the true damage position, and more than 70% fell within the A_0 diffraction limit of 6.7 mm. Notably, symmetry-aware architectures were most effective in challenging boundary regions, reducing both localization error and uncertainty.

5. Conclusions and Future Work

The automated experimental platform successfully generated training data on a scale that rivals computational simulations. The platform achieves orders-of-magnitude throughput gains versus manual testing and consistently captures real-world complexities—rolling-direction anisotropy, bonding variability, and sensor timing differences—that are crucial for model realism. Unlike previous studies that relied on semi-permanent or permanently bonded contact loads,

this platform utilized a temporary coupling method that allowed for the autonomous movement of the contact load across the plate. A systematic couplant study identified 10A-durometer silicone as the optimal choice for autonomous operation, producing strong attenuation of the A0 mode while preserving experimental cleanliness and repeatability. The experimental datasets generated by this platform were instrumental in training the machine learning models described in the published study [20]. By capturing the intricate interactions between Lamb waves and simulated damage, the datasets enabled the development of robust models that achieved over 99% accuracy in detecting damage.

Overall, the framework closes key gaps in SHM: it reduces reliance on overly pristine simulations by supplying large-scale experimental data with realistic noise and variability, and it adapts neural network design to time-series physics rather than repurposing image-centric architectures. The approach is reproducible, scalable, and extensible to other plate materials, sensor layouts, and damage proxies.

Future work will expand contact-load concepts to more crack-like proxies (reflections and scattering), incorporate controlled temperature and environmental variability, and extend fusion to multi-geometry datasets to study transferability across structures.

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